CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 1: Real Numbers

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 HCF and LCM by prime factorisation

1. Find the HCF of 960 and 432. (a) 48 (b) 54 (c) 72 (d) 36 CBSE 2026, 1 mark

SOLUTION

$960 = 2^6 \times 3 \times 5$ and $432 = 2^4 \times 3^3$

HCF = product of the smallest powers of the common primes $= 2^4 \times 3 = 48$

Answer: (a)

2. Let m be the HCF and n the LCM of 98 and 28. Find the value of $n - 7m$. CBSE 2025, 1 mark

SOLUTION

$98 = 2 \times 7^2$ and $28 = 2^2 \times 7$

HCF $m = 2 \times 7 = 14$ and LCM $n = 2^2 \times 7^2 = 196$

$n - 7m = 196 - 98$

Answer: 98

3. Two positive integers are $p = 18a^2b^4$ and $q = 20a^3b^2$, where a and b are primes. What is the LCM of p and q? (a) $2a^2b^2$ (b) $180a^2b^2$ (c) $12a^2b^2$ (d) $180a^3b^4$ CBSE 2024, 1 mark

SOLUTION

$p = 2 \times 3^2 \times a^2 \times b^4$ and $q = 2^2 \times 5 \times a^3 \times b^2$

LCM takes the greatest power of every prime: $2^2 \times 3^2 \times 5 \times a^3 \times b^4 = 180a^3b^4$

Answer: (d)

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4. At a workshop there are 48 French teachers, 80 Hindi teachers and 144 English teachers. Every room must hold the same number of teachers, and all the teachers in a room must teach the same subject. What is the least number of rooms needed? CBSE 2024, 3 marks

SOLUTION

For the fewest rooms, each room must seat the greatest possible number, which is the HCF of 48, 80 and 144.

$48 = 2^4 \times 3$, $80 = 2^4 \times 5$, $144 = 2^4 \times 3^2$, so HCF $= 2^4 = 16$

Number of rooms $= \frac{48}{16} + \frac{80}{16} + \frac{144}{16} = 3 + 5 + 9$

Answer: 17 rooms

5. What is the largest number that leaves remainder 1 when it divides 85 and remainder 2 when it divides 72? CBSE 2023, 2 marks

SOLUTION

The number divides $85 - 1 = 84$ and $72 - 2 = 70$ exactly, so it is the HCF of 84 and 70.

$84 = 2^2 \times 3 \times 7$ and $70 = 2 \times 5 \times 7$

HCF $= 2 \times 7$

Answer: 14

6. Find the HCF of the smallest prime number and the smallest composite number. CBSE 2018, 1 mark

SOLUTION

The smallest prime number is 2 and the smallest composite number is $4 = 2^2$.

Answer: HCF $= 2$

7. Find the HCF and LCM of 404 and 96, and check that HCF $\times$ LCM equals the product of the two numbers. CBSE 2018, 3 marks

SOLUTION

$404 = 2^2 \times 101$ and $96 = 2^5 \times 3$

HCF $= 2^2 = 4$ and LCM $= 2^5 \times 3 \times 101 = 9696$

HCF $\times$ LCM $= 4 \times 9696 = 38784$ and $404 \times 96 = 38784$. The two are equal.

Answer: HCF $= 4$, LCM $= 9696$

2 The Fundamental Theorem of Arithmetic

8. How is the natural number 2 classified? (a) prime (b) composite (c) both prime and composite (d) neither prime nor composite CBSE 2026, 1 mark

SOLUTION

2 has exactly two factors, 1 and 2, so it is prime. It is the only even prime number.

Answer: (a)

9. For a natural number n, what is the last digit of 6^n? (a) 0 (b) 6 (c) 3 (d) 2 CBSE 2026, 1 mark

SOLUTION

$6^1 = 6, 6^2 = 36, 6^3 = 216$. Multiplying a number that ends in 6 by 6 again gives a number that ends in 6.

Answer: (b)

10. For which natural numbers n is $(-1)^n + (-1)^8 = 0$? CBSE 2025, 1 mark

SOLUTION

$(-1)^8 = 1$, because the power is even.

So we need $(-1)^n = -1$, which happens only when the power is odd.

Answer: n is any odd natural number

11. Show that $5 \times 11 \times 17 + 3 \times 11$ is a composite number. CBSE 2024, 2 marks

SOLUTION

$5 \times 11 \times 17 + 3 \times 11 = 11\,(5 \times 17 + 3) = 11 \times 88$

The number has factors other than 1 and itself (for example 11 and 88).

Hence it is a composite number.

3 Irrational numbers and their proofs

12. Show that $\sqrt{3}$ is irrational. CBSE 2026, 3 marks

SOLUTION

Suppose $\sqrt{3}$ is rational. Then $\sqrt{3} = \frac{a}{b}$, where a and b are positive integers with no common factor other than 1.

Squaring, $3b^2 = a^2$. So 3 divides a^2, and therefore 3 divides a. Write $a = 3c$.

Then $3b^2 = 9c^2$, so $b^2 = 3c^2$. So 3 divides b^2, and therefore 3 divides b.

Now 3 is a common factor of a and b, which contradicts the fact that they have no common factor other than 1.

So our assumption is wrong, and $\sqrt{3}$ is irrational.

13. Which of these is a rational number lying between $\sqrt{3}$ and $\sqrt{5}$? (a) $1.4142387954012\ldots$ (b) $2.32\overline{6}$ (c) π (d) 1.857142 CBSE 2025, 1 mark

SOLUTION

$\sqrt{3} \approx 1.732$ and $\sqrt{5} \approx 2.236$, so the number must lie between these two values.

(a) is less than 1.732, and it neither ends nor repeats, so it is irrational. (b) is rational but greater than 2.236. (c) $\pi \approx 3.14$ is irrational and too large.

(d) 1.857142 is a terminating decimal, so it is rational, and $1.732 < 1.857142 < 2.236$.

Answer: (d)

14. Show that $\frac{1}{\sqrt{5}}$ is an irrational number. CBSE 2025, 3 marks

SOLUTION

Suppose $\frac{1}{\sqrt{5}}$ is rational. Then $\frac{1}{\sqrt{5}} = \frac{a}{b}$, where a and b are positive integers with no common factor other than 1.

Cross-multiplying, $b = \sqrt{5}\,a$. Squaring both sides, $b^2 = 5a^2$.

So 5 divides b^2, and therefore 5 divides b. Write $b = 5c$.

Then $25c^2 = 5a^2$, so $a^2 = 5c^2$. So 5 divides a^2, and therefore 5 divides a.

Now 5 is a common factor of a and b. This contradicts the fact that a and b have no common factor other than 1.

So our assumption is wrong, and $\frac{1}{\sqrt{5}}$ is irrational.

15. Given that $\sqrt{3}$ is irrational, prove that $5 - 2\sqrt{3}$ is irrational. CBSE 2024, 2 marks

SOLUTION

Assume, to the contrary, that $5 - 2\sqrt{3}$ is rational. Then $5 - 2\sqrt{3} = \frac{a}{b}$, where a and b are integers and $b \neq 0$.

Rearranging: $\sqrt{3} = \frac{5b - a}{2b}$

Since a and b are integers, the right side is rational, so $\sqrt{3}$ is rational. This contradicts the given fact that $\sqrt{3}$ is irrational.

Hence $5 - 2\sqrt{3}$ is irrational.

16. Show that $\sqrt{5}$ is irrational. CBSE 2023, 3 marks

SOLUTION

Suppose $\sqrt{5}$ is rational. Then $\sqrt{5} = \frac{a}{b}$, where a and b are positive integers with no common factor other than 1.

Squaring, $5b^2 = a^2$. So 5 divides a^2, and therefore 5 divides a. Write $a = 5c$.

Then $5b^2 = 25c^2$, so $b^2 = 5c^2$. So 5 divides b^2, and therefore 5 divides b.

Now 5 is a common factor of a and b, which contradicts the fact that they have no common factor other than 1.

So our assumption is wrong, and $\sqrt{5}$ is irrational.

17. $\triangle ABC$ is right-angled at B with $AB = 2$ cm and $BC = 3$ cm. Assertion (A): The perimeter of $\triangle ABC$ is a rational number. Reason (R): The sum of the squares of two rational numbers is always rational. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2023, 1 mark

SOLUTION

$AC = \sqrt{2^2 + 3^2} = \sqrt{13}$, which is irrational. Perimeter $= 5 + \sqrt{13}$ is irrational, so A is false.

R is true: squares of rational numbers are rational, and so is their sum.

Answer: (d)

18. Write a rational number that lies between $\sqrt{2}$ and $\sqrt{3}$. CBSE 2019, 1 mark

SOLUTION

$\sqrt{2} \approx 1.414$ and $\sqrt{3} \approx 1.732$

Any terminating decimal between these values will do, for example $1.5 = \frac{3}{2}$.

Answer: 1.5 (other answers such as 1.6 or 1.7 are also correct)

19. Show that $\sqrt{2}$ is irrational. CBSE 2019, 3 marks

SOLUTION

Suppose $\sqrt{2}$ is rational. Then $\sqrt{2} = \frac{a}{b}$, where a and b are positive integers with no common factor other than 1.

Squaring, $2b^2 = a^2$. So 2 divides a^2, and therefore 2 divides a. Write $a = 2c$.

Then $2b^2 = 4c^2$, so $b^2 = 2c^2$. So 2 divides b^2, and therefore 2 divides b.

Now 2 is a common factor of a and b, which contradicts the fact that they have no common factor other than 1.

So our assumption is wrong, and $\sqrt{2}$ is irrational.

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20. Given that $\sqrt{2}$ is irrational, prove that $5 + 3\sqrt{2}$ is irrational. CBSE 2018, 2 marks

SOLUTION

Suppose $5 + 3\sqrt{2}$ is rational, equal to $\frac{a}{b}$ where a and b are integers and $b \neq 0$.

Then $3\sqrt{2} = \frac{a}{b} - 5$, so $\sqrt{2} = \frac{a - 5b}{3b}$.

The right side is a ratio of integers, so it is rational. That would make $\sqrt{2}$ rational, which contradicts the given fact.

So $5 + 3\sqrt{2}$ is irrational.

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